Week 21 Homework

Find One New Dataset. Propose One Community Project.

A clear, feasible question is more important than a complicated AI model.

Student: _____ **Date:** _____

Your task

Independently find **one additional dataset not used in the assigned report** and propose **one project of your own** that could help your school, neighborhood, or another clearly identified community. Use the examples' chain:

Data → Question → Decision → Value → Check

You may propose a descriptive analysis, a comparison of options, or a prediction. **No completed model, experiment, intervention, or new code is required.** Estimated effort: **90–120 minutes**. Submit your work at the next class, unless your mentor sets another deadline.

What counts as an additional dataset?

The report uses Boston BERDO building-energy records, NASA POWER solar/weather records, and ISO New England NEMA hourly demand/prices. Your main new dataset must be **different from these**. Another year, location query, copy, or website link for the same dataset does not count. A distinct dataset from the same publisher can count if you explain how it differs. You may use a report dataset as an optional secondary source.

Submit these three items

1. **Dataset card (about one page):** exact title, publisher, source/query link, access date, coverage, row meaning, relevant fields/units, limitations, and reuse/access conditions.
2. **Access evidence:** five genuine sample rows with headers and units, plus a small permitted extract or a screenshot/query record showing where they came from. This evidence can be attached to the dataset card. Do not invent sample values.
3. **Project proposal (1–2 pages):** one question, analysis steps, simple comparison, intended user, decision, possible community value, validation, limitations, and a first-week plan.

You may fill in the following worksheets or submit equivalent headings in your own document. Add sample rows as an attachment; do not include private personal information.

Independence, AI, and safety

Choose the dataset and draft your question **yourself before using AI**. You may later ask AI for wording or skeptical feedback, but personally open the source, verify records, and disclose what changed. Do not use AI to invent evidence or choose your project for you. No public social-media post is required. Use public, non-identifying data; ask a staff mentor before contacting community partners. Do not enter restricted spaces, operate electrical equipment, or collect classmates' personal routines.

Part A: Your Dataset Card

1. Identity and proof of access

- Dataset title and original publisher:

- Exact dataset/download/query link and date accessed:

- How is this different from the three datasets used in the report?

- Evidence attached: file name or screenshot/query description. Could you view/download it without payment, credentials, or an unresolved permission request?

2. Coverage and meaning

- What does **one row** represent? What location, dates, and observation frequency will your subset cover? Approximately how many rows are available in that subset?

3. Relevant fields: preserve the source's names and units

Field name	Meaning	Unit / format	Why you need it

4. Data quality and permission State two concerns such as missing records, coarse geography, delayed publication, different units, or incomplete dates. Record stated reuse terms or say that you could not locate them; identify any permission still needed. A publicly viewable page is not permission to expose personal data.

Part B: Your One-Project Proposal

1. Project title, community, and user Give a short title, a specific place/community, and the person or organization that could use the result. State whether contact is established or only proposed; do not invent a partnership.

2. One main research question Write a bounded question supported by the dataset you actually opened. Specify a place and period. Explain in one sentence how your idea adapts the reasoning of one lecture example without merely copying its question.

3. Minimum analysis plan Write three to five steps: select a subset, check quality/units, make a summary or comparison, and interpret the result. Identify the exact fields used. A spreadsheet or a short Python analysis is sufficient; explain any proposed AI role.

4. Baseline and planned output What simple alternative will you compare against: current practice, a fixed rule, a group median, last year's pattern, or another defensible reference? Describe one planned figure/table, including its axes or columns and units. If predicting, state which information will be available at forecast time and how you will hold out later data.

Part C: Community Value and Reality Check

1. Connect the answer to an action Complete this statement: "[**User**] could use [**my output**] to decide [**specific action**]. This could help [**beneficiary**] by [**mechanism of benefit**]." Explain who is able to make the change and who receives any benefit.

2. Measure usefulness honestly Name one measurable outcome and unit (for example, kWh avoided, actual-contract dollars saved, waiting minutes, or coverage of a service). State the baseline, a reasonable target if appropriate, and how you would obtain evidence. Also name one constraint that must not worsen, such as driver deadlines, comfort, accessibility, or privacy. **A proposed benefit is not an achieved result.**

3. Assumptions, limitations, and a fallback Label at least one assumption separately from observed data. Give two reasons your answer might not generalize or might not justify the proposed action. What important extra information is missing? If it is unavailable, how could you narrow the question to something the public data can answer?

4. A realistic first seven days List three small actions, an expected output, and any permission/contact needed. For example: confirm coverage, clean a small subset, and draft one comparison chart. Do not promise an installation or intervention without adult authorization.

5. AI-use statement Write "No AI used," or name the tool, describe its limited role, and say what you checked or changed. Confirm that you independently chose the dataset and drafted the question first.

Discovery Guide

Possible starting points—you still choose the dataset

- NOAA Climate Data Online: look for daily weather observations at a relevant station and a defined past period. A possible question concerns seasonal patterns; a distant station does not directly measure conditions on your school grounds.
- EPA AirData daily download: select a pollutant, year, geography, and monitor. Historical pollution analysis is not a current-condition alert or medical recommendation.
- Your city/county's official open-data catalog: search for a *specific* dataset on trees, transit, waste, public buildings, or another issue you care about. Verify local availability rather than assuming it exists.

Record the **actual dataset/query link**, not only the portal homepage. Check observation dates, field definitions, units, missing values, and access/reuse conditions. Discovery entry pages were checked on October 4, 2026; individual local coverage and access may change.

If your search gets stuck

After about 20 minutes with an inaccessible or unsuitable source, record the problem and switch to another one. Prefer a small public table you can actually inspect. If local data are unavailable, choose the nearest relevant public dataset and clearly state its geographic limitation, or narrow the question. An unverified dataset wishlist is not a completed dataset card.
Lecture reference: `prism-uploads/Community_Energy_AI_Report.pdf`, especially pp. 17–19 for deliverables, data sources, and methods.